WEEK 1

Big Data and Data Mining

Defining Big Data: Big data refers to information that is high-volume, high-velocity, and high-variety, requiring new cost-effective forms of processing. This is commonly expanded into the "5 Vs": Volume (scale exceeding single-machine capacity), Velocity (the speed of data generation and processing), Variety (structured, semi-structured, and unstructured formats), Veracity (data trustworthiness, accuracy, and completeness), and Value (the business benefit derived from analyzing the data).

Data Mining: This is the process of discovering patterns in large datasets that a person could not find by reading the data directly. Common analytical techniques include clustering (grouping similar behavior), classification/prediction (forecasting outcomes), association rules (finding items frequently bought together), and anomaly detection (flagging unusual records)

System Types: The lecture distinguishes between Operational systems (OLTP), which handle high volumes of day-to-day short transactions, and Analytical systems (OLAP), which handle complex queries on historical data to support business decision-making.

Artificial Intelligence

Defining AI & Intelligence: AI enables machines to simulate human learning, problem-solving, and adaptability. To understand how to simulate this, the lecture covers various intelligence theories, including Spearman's general intelligence, Thurstone's primary mental abilities, Gardner's multiple intelligences, Fluid vs. Crystallized intelligence, and Emotional intelligence.

Core Concepts: AI is broadly categorized into Weak (Narrow) AI, Strong (General) AI, and Super AI. The lecture highlights the difference between traditional programming, which applies manual rules to data to generate answers, and machine learning, which uses data and correct answers to independently learn the underlying rules. The broader field of AI acts as an umbrella encompassing machine learning, deep learning, and neural networks.

Building Intelligent Entities: The goals of AI development are modeled across four quadrants: Thinking Humanly (cognitive science and information processing), Acting Humanly (natural language processing and the Turing Test), Thinking Rationally, and Acting Rationally (autonomous agents pursuing goals).

Philosophy & History: The lecture discusses John Searle's "Chinese Room" argument, which challenges the concept of Strong AI by arguing that mindlessly manipulating symbols (syntax) does not equate to genuine understanding or consciousness (semantics). It also traces AI's history from the foundational 1956 Dartmouth workshop, through periods of reduced funding known as the "AI Winter," to the modern resurgence of AI driven by massive increases in data, computing power, and advanced algorithms.

Applied Machine Learning

AI, ML, and Deep Learning Concepts: Artificial Intelligence involves machines performing tasks that require human intelligence, Machine Learning consists of systems that learn patterns from data rather than explicit rules, and Deep Learning uses multi-layer neural networks to learn from raw data. ML relies on three primary components: data (the recorded examples), algorithms (the learning methods), and models (the representation of what has been learned).

Types of Machine Learning: The field is categorized into Supervised Learning (using labeled data), Unsupervised Learning (finding patterns in unlabeled data), and Reinforcement Learning (learning through rewards and penalties). Modern hybrid approaches also include Semi-Supervised, Self-Supervised, and Online Learning.

The ML Process and Challenges: Developing an ML system is an iterative seven-step process: problem definition, data collection, data preprocessing, model selection, training, evaluation, and deployment. Practitioners must navigate technical challenges like poor data quality and overfitting, alongside legal constraints like GDPR, ethical concerns regarding bias, and moral issues like deepfakes.

Tools and Roadmap: The course relies heavily on Python and its ecosystem, including NumPy, Pandas, Matplotlib, Scikit-learn, and PyTorch, as well as big data frameworks like PySpark and Hadoop. The 12-week module will progress through data visualization, supervised and unsupervised learning, Natural Language Processing (NLP), and reinforcement learning, culminating in a capstone project.